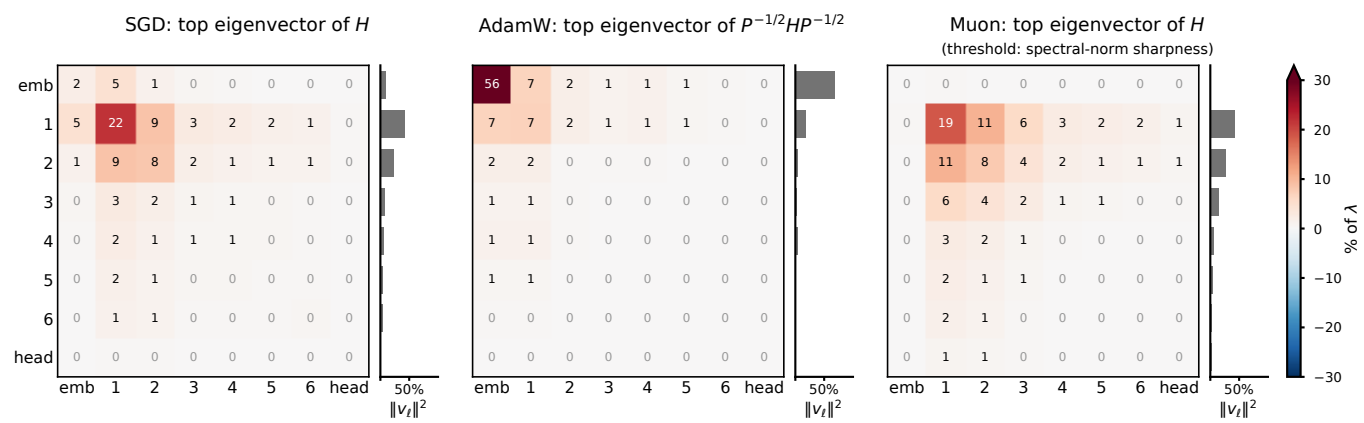
SGD: top eigenvector of H
AdamW: top eigenvector of $P^{-1/2}HP^{-1/2}$
Muon: top eigenvector of H
(threshold: spectral-norm sharpness)
emb 1 2 3 4 5 6 head
50% $\|v_\ell\|^2$
% of λ
30 20 10 0 −10 −20 −30